

Name			
USN			
Date	Conduction	Record	Viva
Allotted Marks			
Obtained Marks			

EXPERIMENT 7

Use Hive to create, alter, and drop databases, tables, views, functions, and indexes

Aim: To create, alter, and drop databases, tables, views using Hive in Apache Ambari

Theory:

Apache Ambari is a **management and monitoring tool** for Hadoop clusters. It provides a web-based UI and RESTful APIs to deploy, configure, and manage services like **Hive, HDFS, YARN, and more**. Apache Hive is a **data warehouse** infrastructure built on top of Hadoop that allows users to query and manage large datasets using **HiveQL** (similar to SQL). It converts queries into MapReduce, Tez, or Spark jobs for execution on **HDFS (Hadoop Distributed File System)**.

Using **Apache Hive on Ambari** makes it much easier to deploy, configure, monitor, and optimize Hive in a Hadoop cluster. It eliminates manual configurations and allows administrators to manage everything through an intuitive web interface.

Procedure:

7.(a) To create, alter, and drop databases

1. Start the Hortonworks virtualbox & wait for the below screen

```
Hortonworks HDP Sandbox
https://hortonworks.com/products/sandbox

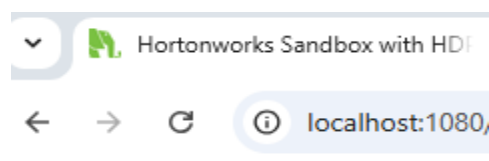
To quickly get started with the Hortonworks Sandbox, follow this tutorial:
https://hortonworks.com/tutorial/hadoop-tutorial-getting-started-with-hdp/

To initiate your Hortonworks Sandbox session, open a browser to this address:

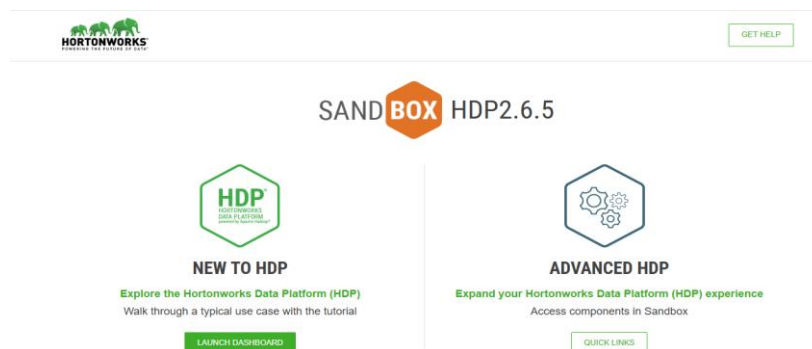
For VirtualBox:
Welcome screen: http://localhost:1080
SSH: http://localhost:4200

For VMWare:
Welcome screen: http://10.0.2.15:1080
SSH: http://10.0.2.15:4200
```

2. go to the web browser and type localhost:1080 and press enter

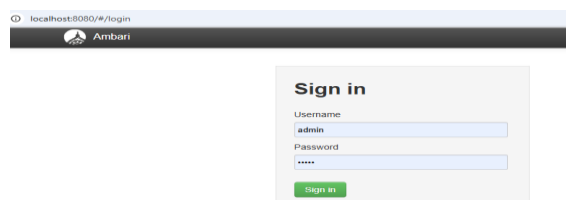


You will see the below screen



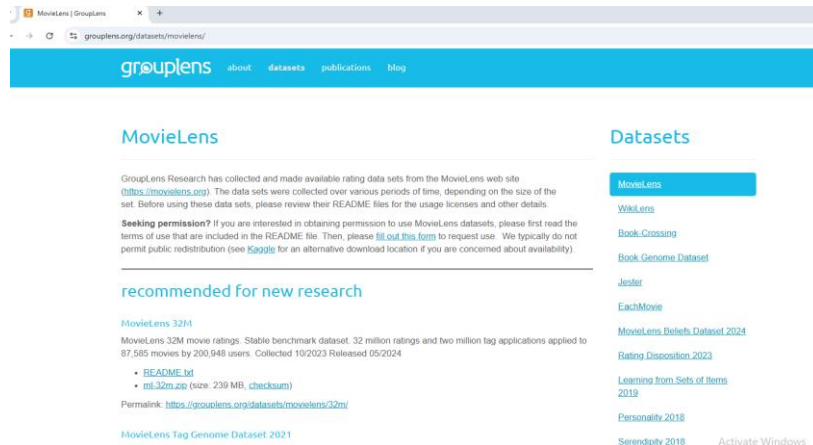
Then click Launch Dashboard

3. Next, you will be navigated to Ambari screen, Type username: **admin** & password: **admin**



4. As we will deal with movie datasets, go to web browser and type movielens.com .

Scroll through older datasets & download ml-100k.zip file & unzip the data file. Find the files u.data & u.item.



older datasets

MovieLens 100K Dataset

MovieLens 100K movie ratings. Stable benchmark dataset. 100,000 ratings from 1000 users on 1700 movies. Released 4/1998.

- [README.txt](#)
- [ml-100k.zip](#) (size: 5 MB, [checksum](#))
- [Index of unzipped files](#)

Permalink: <https://grouplens.org/datasets/movielens/100k/>

ml-100k

ml-100k

u.data	14-03-2025 10:24	DATA File	1,933 KB
u.genre	14-03-2025 10:24	GENRE File	1 KB
u.info	14-03-2025 10:24	INFO File	1 KB
u.item	14-03-2025 10:24	ITEM File	221 KB

5. Now go to Ambari dashboard, grid view & select Hive view. Select the query & go to query editor .

type the following commands

here we are trying to create, alter & drop the database ; database_test

To Create

To Alter

To drop/ delete

7.(b) To create, alter, and drop tables

7.(c) To create, alter, and drop VIEWS

Result**To Drop/Delet****Inference:**